

Ph.D. candidate in the Department of Computer Science, University of Virginia, Charlottesville, Virginia, United States. My research focuses on enhancing **human-computer/human-AI interaction** through multimodal interaction, haptic devices, computer vision, and robotics. I build multimodal interactive systems that connect physical and digital experiences.

RESEARCH

- Teleoperated Robotic Hands with Haptic Feedback for Precise Remote ControlMay 2025 — Now
 - Developing a VLA-based teleoperation system that combines vision and language inputs for precise robotic hand control.
 - Integrating real-time haptic feedback to improve awareness, responsiveness, and precision during remote manipulation tasks.
- Instant Tangible User Interfaces for Wearable Displays Using Everyday ObjectsMar 2025 — Apr 2026
 - Turned everyday objects into tangible inputs for wearable displays, supporting multimodal interactions.
 - Built a transformer-based vision pipeline for hand-object detection (90.0% accuracy) and input inference (97.5% accuracy).
- A Dynamic Shape Display for Enhanced On-Surface Interactions in VRFeb 2024 — May 2025
 - Built a foldable shape-changing device that uses shape approximation and visuo-haptic illusions to render VR surfaces.
 - Studies showed the system improved realism, control, and performance in 3D modeling tasks over tablet and mid-air input.
- Efficient and Ergonomic Multimodal Interactions through Self-Haptics in XRJan 2023 — Apr 2024
 - Created a hand touchpad supporting efficient AR/VR interactions using an ML model with 92% touch detection accuracy.
 - Built a system enabling precise and comfortable hand-gaze interaction for multi-window productivity tasks.
- Effects of Different Haptic Feedback on Precise Bimanual Interactions in VRMar 2021 — Nov 2022
 - Compared visual-only feedback, tactile+kinesthetic feedback, and physical feedback in various precision-required tasks in VR.
 - The physical surface improved selection accuracy, tracing precision, and sketch quality over other haptic conditions.

EXPERIENCE

- Research InternshipMay 2026 — Now

Autodesk | Future AI TeamSan Francisco, CA, USA

 - Building a real-time human-AI co-design workflow for CAD using multimodal interaction and multi-agent orchestration.
- Research AssistantSep 2020 — Now

UVA Ultimate User Interface Lab | With Prof. Seongkook HeoCharlottesville, VA, USA

 - System design/prototyping, computer vision, machine learning, signal processing, user studies, data analysis in HCI.

EDUCATION

- University of Virginia, Master/Ph.D., USA GPA: 3.89/4.00Sep 2018 — Now
- Zhejiang University, Bachelor, China, GPA: 3.70/4.00Sep 2014 — Jun 2018

TECHNICAL SKILLS

Hardware	Interaction Technologies, Haptic Systems, Circuit/PCB Design, 3D Modeling and Printing
Software	Unity, Meta Oculus, Autodesk Fusion 360, Arduino, OptiTrack Tracking, Generative AI
Programming	Python, PyTorch, Java, JavaScript, C#, R, C/C++, ROS, Matlab

PUBLICATIONS

- Ying, W.**, Kim, Y., Rahman, A., *et al.* Redirected Pinch: Efficient and Comfortable Bare-Hand Interaction for 2D Windows in VR. *Proceedings of the CHI Conference on Human Factors in Computing Systems* (April 13-17, 2026).
- Ying, W.** & Heo, S. Physical surfaces make touch interactions in virtual reality precise, efficient, and bimanual. *International Journal of Human-Computer Studies* (June 9, 2026).
- Rahman, A., **Ying, W.**, Azim, M. A. R., *et al.* Mediating Ad-Hoc Bystander-VR User Interruptions Through Proactive Proxies. *Proceedings of the CHI Conference on Human Factors in Computing Systems* (April 13-17, 2026).
- Ying, W.** & Heo, S. Enhancing VR Sketching with a Dynamic Shape Display (**Best Paper Honorable Mention**). *ACM Symposium on Virtual Reality Software and Technology* (October 9-11, 2024).
- Zhang, P., **Ying, W.**, Riggs, S., *et al.* MoiréTag: A Low-cost Tag for High-precision Tangible Interactions Without Active Components (**Honorable Mention**). *ACM Interactive Surfaces and Spaces Conference* (October 27-30, 2024).
- Hu, E., Grønbaek, J. E. S., **Ying, W.**, *et al.* ThingShare: Ad-Hoc Digital Copies of Physical Objects for Sharing Things in Video Meetings. *Proceedings of the CHI Conference on Human Factors in Computing Systems* (April 23-28, 2023).
- Hildebrandt, C., **Ying, W.**, Heo, S., *et al.* Mimicking Real Forces on a Drone Through a Haptic Suit to Enable Cost-Effective Validation. *IEEE International Conference on Robotics and Automation* (May 29-June 2, 2023).